10 MHz Reference and Timebase

CP-FTMW spectrometer · Tabor Proteus P9482D + Tektronix MSO64B · as built 14 October 2026

The AWG and the oscilloscope each contain their own clock. Left alone those clocks run at slightly different rates, and each instrument is entitled to be a few parts per million off nominal. That mismatch costs two separate things, worth separating because they differ in size, in symptom, and in cure.

- **Absolute frequency accuracy** is set by the AWG's reference alone. Everything the AWG emits is clocked from it, and the multiplier chain carries the fractional error straight up to 350 GHz.
- **Shot-to-shot coherence** is set by the difference between the AWG's clock and the scope's. It limits how long an FID can be coherently integrated and how many shots can be averaged.

One external reference feeding both instruments removes both at the source. This document records why, what was built, what was measured, and the rules for operating it.

1. Failure mode one — absolute frequency error

The AWG's output frequency is proportional to its reference, so a fractional reference error δ appears as the same fractional error on every frequency the instrument produces. Multiplication by 24 scales the absolute error with the signal and leaves the fraction untouched:

```
f_error  =  f_output  x  delta
```

Reference in use	delta	Error at 350 GHz
Proteus internal TCXO, best case	1 ppm	350 kHz
Proteus internal TCXO, worst case	2.5 ppm	875 kHz
Wenzel OCXO	0.002 ppm	700 Hz

A few hundred kilohertz does not blur a spectrum. It shifts every line by the same amount while leaving the data looking entirely healthy, which is the dangerous kind of error: it survives every internal consistency check and only surfaces when a line is matched against a catalogue. This is the single strongest argument for locking the AWG, and it is independent of averaging.

2. Failure mode two — timebase stretch

Define the relative error between the two instruments:

```
delta  =  ( f_scope - f_awg ) / f_nominal
```

The scope believes its own sample interval is exactly nominal, so it reports the AWG's time axis scaled by (1 + delta). Over a record of length T:

```
timing error   dt    =  delta x T
phase error    dphi  =  360 x f x delta x T      [degrees]
```

Evaluated at delta = 2.5 ppm, the Proteus TCXO worst case. The scope's own 20 ppb is negligible beside it.

Signal	Record length	dt	Phase at end of record
4 GHz, directly digitised	10 µs	25 ps	36°
800 MHz IF	10 µs	25 ps	7.2°
800 MHz IF	100 µs	250 ps	72°

Static stretch versus drift

The distinction matters and is easy to miss. If delta is constant, every shot is distorted identically. That is a systematic scale error: it biases the measured frequency by delta (2 kHz on an 800 MHz IF at 2.5 ppm) and it is precisely what `align_and_average()` cannot remove, since that routine corrects a time offset and not a stretch. It does not, however, smear coherent averaging.

What smears averaging is delta drifting between shots, as a TCXO does with temperature. The phase error is then zero at the trigger and grows along the record, so the tail of the FID decoheres while the head stays put. That is not a uniform amplitude loss — it is line broadening.

The useful figure of merit

Express it as the integration time at which shot-to-shot phase scatter reaches about one radian:

```
t_coh  =  1 / ( 2*pi x f_IF x delta )

delta = 2.5 ppm,  f_IF = 800 MHz   ->   t_coh =  80 us
delta = 2   ppb,  f_IF = 800 MHz   ->   t_coh = 100 ms
```

A factor of 1250. On the internal TCXO, clock mismatch sits right at the length of a typical FID and quietly sets the resolution floor. On the OCXO it stops being anywhere near the limiting term.

3. The chain as built

```
  Wenzel 501-27501-11   10 MHz OCXO
  +14.2 dBm measured, 15 V, runs 24/7
       |
       |  SMA
       v
  Mini-Circuits ZX10R-2-183-S+
  2-way resistive, 6.3 dB loss,
  ~11 dB isolation at 10 MHz
       |
       +------------------------------+
       |                              |
    arm 1                             arm 2
    (keep this length)                (any length)
       |                              |
    2.92 Vpp                          1.95 Vpp
       |                              |
       v                              [ 6 dB pad ]
  SCOPE Ref In, rear BNC              |
  745 ohm, unterminated               0.98 Vpp
  LOCKED, External 1 ppm              |
                                      v
                                      AWG REF IN, SMP
                                      50 ohm AC coupled
                                      [ still to do ]
```

Measurements taken 14 October 2026 on MSO64B CH3 at 50 ohm, 100 averages.

What was measured and what follows from it

Condition	Pk-Pk	Amp	RMS	Vpp used
Operating: arm 1 on Ref In	1.963 V	1.915 V	0.6866 V	1.95
Diagnostic: arm 1 into 50 ohm	1.576 V	1.56 V	0.5526 V	1.563

RMS × 2.828 reproduces the pk-pk value to within 1% in both cases, which confirms a clean sine rather than a squared-up clock — a square wave would have given a 41% discrepancy. Those two readings determine both unknowns:

```
OCXO output    P = 2.5 x 1.563^2 = 6.11 mW = +7.86 dBm at the port
                 + 6.3 dB splitter loss      = +14.2 dBm
                 (spec +13 +/- 2 dBm, so a hot but in-spec unit)

Isolation      ratio = 1.942 / 1.563 = 1.243
               Iso   = -20 log10( 0.243 / 0.875 ) = 11 dB
               (datasheet band 8-13 dB; this is the in-phase bound)
```

The hot oscillator is why the operating reading landed above the level predicted from the nominal datasheet figures.

4. Level budget, and why the two arms differ

Each source port behaves as one voltage source behind 50 ohm, so the load forms a divider with that internal resistance. A matched 50 ohm load therefore receives exactly half the source EMF; a high-impedance load receives nearly all of it.

```
V_load  =  V_th  x  R_L / ( R_L + R_th )

R_L =  50 ohm   ->  0.50 x V_th         (a termination)
R_L = 745 ohm   ->  0.94 x V_th         (a sensing input)

ratio between them = 2 x 745 / 795 = 1.87
```

The scope's Ref In is deliberately 745 ohm because it senses voltage rather than absorbing power; that is why it must be left unterminated. Fitting a 50 ohm feedthrough would drop it from 2.92 Vpp to 1.56 Vpp, below its 1.5 Vpp typical sensitivity, and the lock could fail.

Node	Impedance	Requirement	Delivered
Scope Ref In (rear BNC)	745 ohm || 18.5 pF	1.5 to 7 Vpp	2.92 Vpp
AWG REF IN (SMP)	50 ohm, AC coupled	0.6 to 1.7 Vpp	0.98 Vpp

Useful conversion, into 50 ohm: `Vpp = 0.632 x sqrt(P_mW)`, so 0 dBm is 632 mVpp.

> **Cable rule**
>
> Arm 1's length sets the phase of the reflection coming back off the 745 ohm input, and therefore where in the interference band the arm 2 level lands. Do not re-route arm 1. Arm 2's length is irrelevant — both the direct and the reflected path traverse it, so it cancels. The 0.98 Vpp figure carries over to the AWG whatever cable is used on arm 2.

5. Operating rules

- **Leave the OCXO powered continuously.** See section 6.
- **Scope:** Settings bar → double-tap the Acquisition badge → Timebase Reference Source → **External, 1 ppm**. Not the 1000 ppm option: the wide setting will lock to an off-frequency reference and silently treat it as exactly 10 MHz, biasing every time measurement.
- **The scope setting is nonvolatile but File → Default Setup clears it.** Re-select External after any default, and after recalling a saved setup.
- **AWG:** send `:ROSC:FREQ 10M` before `:ROSC:SOUR EXT`. The frequency parameter defaults to 100M, so the source command alone will try to lock to 100 MHz and fail.
- **There is no lock-status query on the Proteus.** Verify empirically: with the scope disciplined, put out a CW tone and watch its phase over a long acquisition. Locked, it sits still; unlocked, roughly 1 ppm of drift is unmistakable.
- **Never leave a splitter port open.** An open port reflects fully and shifts the level on the other arm by up to 2 dB. Terminate anything unused in 50 ohm.
- **Do not terminate the scope's Ref In.** It is a 745 ohm bridging input and is meant to be driven unterminated.

Loss of reference

The Proteus manual states that in external mode "an external reference must be connected to the Proteus for it to continue with its normal operation", which reads as a hard dependency rather than graceful degradation. The scope's behaviour on reference loss has not yet been tested. Pull the cable once deliberately and find out what the failure looks like, rather than meeting it mid-experiment.

6. Power cycling and aging

Continuous operation makes an OCXO age more slowly, not faster, which is the opposite of the intuition that applies to mechanical equipment.

Crystal aging is driven by mass transfer on the crystal surface, stress relaxation in the mount, and outgassing inside the enclosure. Those are driven by temperature changes rather than by running time. An oscillator held at constant oven temperature settles onto a smooth, decaying aging curve. Each power cycle thermally cycles the crystal and its mount, perturbs that curve, and restarts a period of faster aging.

How many cycles is too many?

There is no cycle budget to exhaust. OCXOs are continuous-duty parts with no rated cycle count, and commercial instruments containing them are switched on and off daily for decades without issue. The cost of a power cycle is not wear — it is **retrace**, the failure of the oscillator to return to exactly the frequency it held before.

Effect of one power cycle	Typical magnitude	At 350 GHz
Retrace, immediately after warm-up	1 to 10 ppb	0.35 to 3.5 kHz
Residual after 24 to 48 h settling	approaching spec	sub-kHz
Cumulative damage	none	—

The point to notice is that a 1 to 10 ppb retrace is as large as, or larger than, this unit's entire ±2 ppb temperature specification. A power cycle does not damage the oscillator, but it does throw away the grade of oscillator that was paid for, for a day or so afterwards.

Practical rule. Cycle it freely when you must — moving the rack, maintenance, building power work. Simply do not expect ppb-level accuracy for 24 to 48 hours afterwards, and do not schedule a frequency-critical measurement into that window. For routine daily operation, leave it on; there is nothing to gain by switching it off and a day of settling to lose by doing so.

A second, practical reason: the oven needs about five minutes to reach temperature, while the instruments boot in seconds. A rack that powers up together will have the AWG and scope looking for a reference that is not ready yet.

Retrace figures here are typical of good OCXOs; the specific figure for this unit should be read off its own datasheet, which was not available at the time of writing.

7. Open items

- Fit the 6 dB pad and run arm 2 to the AWG REF IN.
- Set the AWG to external reference and verify the lock empirically.
- Add a reference-lock guard to the control software: check both instruments report an external reference at connect time, and warn loudly otherwise. A silent fallback to internal is invisible on screen and costs 350 to 875 kHz of line position.
- Test deliberately what each instrument does when the reference disappears.
- Separate but related: marker output from the AWG to the scope's AUX TRIG IN, which is what makes coherent averaging work once the clocks are shared.